

*IEEE VTC-Spring 2026: The 3rd Workshop on
Synergizing Digital Twins and Pervasive Intelligence for Next-Generation IoT/IoV*

I. Workshop Organizer

Workshop General Co-chairs

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Workshop TPC Co-chairs

Qiang (John) Ye, of Calgary, Canada (qiang.ye@ucalgary.ca)
Qihao Li, Jilin University, China (qihaol@jlu.edu.cn)
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Workshop Publicity Co-chair

Jie Gao, Carleton University, Canada (jie.gao6@carleton.ca)
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Omar Alhussein, Khalifa University, UAE (omar.alhussein@ku.ac.ae)

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Invited Keynote Speakers

Prof. Yansha Deng, King's College London, UK (yansha.deng@kcl.ac.uk), (to be confirmed)
Prof. Berk Canberk, Edinburgh Napier University, UK (B.Canberk@napier.ac.uk) (to be confirmed)

Website: <https://events.vtsociety.org/vtc2026-spring/conference-sessions/workshops-available/synergizing-digital-twins-and-pervasive-intelligence-for-next-generation-IoT-IoV>

II. Biographies of Workshop Organizers

Trung Q. Duong (Fellow of IEEE, Fellow of EIC, Fellow of IET, Fellow of CAE, Fellow of AAIA) is a Canada Excellence Research Chair (CERC) and a Full Professor at Memorial University of Newfoundland, Canada. He is also an Adjunct Professor at Queen's University Belfast, UK. He has received the two prestigious awards from the UK Royal Academy of Engineering (RAEng), including: 1) RAEng Research Chair and 2) RAEng Research Fellow. In 2017, he was awarded the Newton Prize from the UK government. He is currently serving as the Editor-in-Chief (EiC) of IEEE Communications Survey & Tutorials (IF=46.7 and CiteScore=86.2). He is an IEEE ComSoc Distinguished Lecturer. He has delivered more than 50 keynote speeches and more than 100 invited lectures, talks worldwide. His current research interests include quantum machine learning and quantum optimization, wireless communications and networks, physical layer security and privacy, machine learning and realtime optimisation, disaster management, smart agriculture, healthcare, air-quality, finance, climate change, and pharmacy.

Tom H. Luan (Fellow, IEEE) received his B.E. degree from Xi'an Jiaotong University, China, his M.S. degree from the Hong Kong University of Science and Technology, and his Ph.D. degree from the University of Waterloo, Canada, all in Electrical and Computer Engineering. From 2013 to 2017, he served as a Lecturer in Mobile and Apps at Deakin University, Australia. From 2017 to 2022, he was a Professor in the School of Cyber Engineering at Xidian University. Since 2022, he has been a Professor in the School of Cyber Science and Engineering at Xi'an Jiaotong University, China. Dr. Luan's research interests include the optimization and protocol design of the Metaverse, digital twin networks, AI-agent networks, as well as the security and performance evaluation of vehicular networks and edge computing systems. He has authored over 270 peer-reviewed papers in top journals and conferences, such as IEEE/ACM Transactions on Networking (TON), IEEE Transactions on Mobile Computing (TMC), IEEE Transactions on Multimedia (TMM), IEEE Transactions on Vehicular Technology (TVT), and IEEE INFOCOM. He received the IEEE VTS Best Land Transportation Paper Award in 2017 and the Best Paper Award at IEEE VTC 2023 fall.

Michael Cheffena is a Professor at NTNU Gjøvik, Faculty of Engineering. He conducts research and teaching within the field of radio communications, signal processing for wireless communication systems, and sensors. His research activities are mainly centered on issues related to antenna and wave propagation for wireless communication systems and their system aspects. In recent years, he has also been interested in: i) Antennas and antenna sensors; ii) Millimeter wave and massive multiple-input and multiple-output (MIMO) techniques; iii) Cooperative communications; iv) Physical-layer security of wireless systems; v) Game theory for optimizing the security of wireless systems; and vi) Machine learning techniques for solving problems in wireless systems. He has a Ph.D in telecommunications (2008), M.Sc in signal processing for communication systems (2005) and B.Sc in computer science (2003).

Mingzhe Chen (Senior Member, IEEE) is currently an assistant professor with the Department of Electrical and Computer Engineering and the knight foundation chair in Data Science and AI in the Frost Institute of Data Science and Computing with the University of Miami, Coral Gables, FL, USA. His research interests include federated learning, reinforcement learning, virtual reality, unmanned aerial vehicles, and Internet of Things. He was the recipient of the four IEEE Communication Society journal paper awards including IEEE Marconi Prize Paper Award in Wireless Communications in 2023, Young Author Best Paper Award in 2021 and 2023, and Fred W. Ellersick Prize Award in 2022, and four conference best paper awards with ICCCN in 2023, IEEE WCNC in 2021, IEEE ICC in 2020, and IEEE GLOBECOM in 2020. He is an associate editor for IEEE Transactions on Mobile Computing, IEEE Transactions on Communications, IEEE Wireless Communications Letters, IEEE Transactions on Green Communications and Networking, and IEEE Transactions on Machine Learning in Communications and Networking.

Qihao Li is an Associate Professor with the College of Communication Engineering, Jilin University, China. He received the M.Sc. degree in Information and Communication Technology from University of Agder, Norway, in 2013 and Ph.D. degree in the Department of Electrical and Computer Engineering from the University of Oslo, Norway, in 2019. In 2016, he was a visiting researcher at the Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, ON, Canada. In 2020, he was a postdoctoral fellow with the Department of Electrical and Computer Engineering from University of Waterloo, ON, Canada. His current research focuses on industrial IoT, digital twin, optimal control and optimization, wireless network security and localization. He serves as Associate Editor of IEEE Internet of Things Journals, and Guest Editor of MPDI Future Internet Journals. He served as the Publication, Program Co-chairs for IEEE VTC' 25-Spring, IEEE Globecom' 24, IEEE CIC/ICCC'23/24, and members of TPC for IEEE Globecom'19-24, IEEE ICC'19-24, IEEE CIC ICC'17-24, EuCAP'2019, BDEC-SmartCity'18.

Qiang (John) Ye received the PhD degree in Electrical and Computer Engineering from the University of Waterloo, ON, Canada, in 2016. Since Sept. 2023, he has been an Assistant Professor with the Department of Electrical and Software Engineering, Schulich School of Engineering, University of Calgary, AB, Canada. Before joining UCalgary, he worked as an Assistant Professor with the Memorial University of Newfoundland, NL, Canada from 2021 to 2023 and with the Minnesota State University, Mankato, USA, from 2019 to 2021, respectively. He was with the Department of Electrical and Computer Engineering, University of Waterloo as a Postdoctoral Fellow and then a Research Associate from 2016 to 2019. He has published over 90 research articles on top-ranked journals and conference proceedings. He is/was the General, Publication, Program Co-chairs for different reputable international conferences and workshops (e.g., IEEE INFOCOM, GLOBECOM, VTC, ICC, ICCT, WISEE, SWC). He also serves/served as the IEEE Vehicular Technology Society (VTS) Region 7 Chapter Coordinator in 2024, the IEEE Communications Society (ComSoc) Southern Alberta Chapter Vice Chair from 2024, and the VTS Regions 1-7 Chapters Coordinator from 2022 to 2023. He is the leading co-chair of a special interest group (SIG) in the IEEE ComSoc - Internet of Things, Ad Hoc & Sensor Networks (IoT-AHSN) Technical Committee. Dr. Ye serves as an Associate Editor for prestigious IEEE journals, such as IEEE Internet of Things Journal (IoT-J), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Cognitive Communications and Networking (TCCN), and IEEE Open Journal of the Communications Society (OJ-COMS). He received the Best Paper Award in the IEEE/CIC International Conference on Communications in China (ICCC) in 2024, the IEEE Open Journal of the Communications Society Exemplary Editor Award in 2025, the IEEE Transactions on Cognitive Communications and Networking Exemplary Editor Award in 2023, the Early Career Research Excellence Award, Schulich School of Engineering, University of Calgary, in 2024. He has been named among the World's Top 2% Scientists in 2023-2025 (by Stanford/Elsevier). He has been selected as an IEEE Communication Society Distinguished Lecturer for the class of 2025-2026. Dr. Ye is a Senior Member of IEEE.

Mushu Li is an Assistant Professor in the Department of Computer Science and Engineering at Lehigh University, USA. She received the Ph.D. degree in Electrical and Computer Engineering from the University of Waterloo, ON, Canada, in 2021. She was a Postdoctoral Fellow at Toronto Metropolitan University (from 2022 to 2024) and the University of Waterloo (from 2021 to 2022). Her research interests include machine learning for communications and networking, cloud and multi-access edge computing, Intelligent Internet of Things (IoT), and next-generation (6G) wireless networks. She has served as an associate editor for *Peer-to-Peer Networking and Applications* since 2023. Dr. Li received the Best

Land Transportation Paper Award from the IEEE Vehicular Technology Society in 2024, the NSERC Postdoctoral Fellowship in 2022, and the NSERC Canada Graduate Scholarship in 2018.

Nan Chen is an Assistant Professor with the Department of Electrical and Computer Engineering, Tennessee Tech University. She received the PhD degree in the Department of Electrical and Computer Engineering from the University of Waterloo, Canada, in 2019. Her research interests include digital twin platform development of smart cities, machine learning application in smart grid and vehicular network, resource allocation in electric vehicle-integrated smart grid and vehicular network. She is an associate editor for peer-to-peer networking and applications (PPNA).

III Scope and Topics of the Workshop

As the Internet of Things (IoT) and the Internet of Vehicles (IoV) evolve toward highly autonomous, hyper-connected, and intelligence-driven ecosystems, future networking, computing, and control infrastructures must support massive connectivity, high mobility, heterogeneous services, and ultra-reliable low-latency communications. These requirements span diverse IoT/IoV scenarios, including connected and autonomous vehicles, intelligent transportation systems, roadside and aerial sensing, smart infrastructure, and edge-cloud collaborative environments. However, such systems face fundamental challenges arising from dynamic large-scale topologies, time-varying wireless conditions, intermittent connectivity, spectrum scarcity, ultra-dense deployments, and stringent safety and latency demands for mission-critical applications. Digital twins (DTs) have emerged as a powerful paradigm for managing the complexity of next-generation IoT/IoV systems by providing virtual representations of physical entities, networks, and processes. By continuously reflecting system states and environmental dynamics, DTs enable real-time monitoring, predictive analysis, and what-if evaluation across physical and virtual domains. Meanwhile, pervasive intelligence powered by artificial intelligence (AI) and machine learning introduces adaptive learning, reasoning, and autonomous decision-making capabilities distributed across devices, vehicles, edge nodes, and cloud platforms. The synergy between DTs and AI is particularly transformative. DTs provide rich, context-aware data and models for AI-driven training and inference, while AI enhances DTs through model adaptation, behavior prediction, anomaly detection, and automated control. This tight DT-AI coupling enables proactive and autonomous IoT/IoV operations, including intelligent resource management, context-aware routing and scheduling, adaptive network slicing, cooperative perception and control, predictive maintenance, and resilient self-optimization, key enablers for future 6G-enabled IoT/IoV systems.

This workshop aims to bring together researchers and practitioners to explore advanced theories, system designs, and experimental implementations that synergize digital twins and pervasive intelligence for next-generation IoT/IoV. Emphasis will be placed on AI-native architectures, edge-cloud intelligence, 6G-enabled platforms, and practical testbeds, fostering interdisciplinary collaboration and accelerating innovation across IoT, IoV, and intelligent transportation ecosystems.

IV. What are the learning objectives? (500 word max)

This workshop aims to equip participants with a comprehensive understanding of the transformative role of **DTs and pervasive AI** in enabling next-generation IoT and IoV systems. The primary learning objective is to elucidate the fundamental principles, system architectures, and emerging methodologies for designing, operating, and optimizing large-scale, heterogeneous, and highly dynamic IoT/IoV environments that demand ultra-reliable, low-latency, and intelligent operation. These include connected and autonomous vehicles, intelligent transportation systems, cooperative sensing, and edge-cloud collaborative networks envisioned for 6G. Participants will gain insights into how DTs provide virtual representations of physical devices, vehicles, networks, and environments by continuously reflecting system states, mobility patterns, communication conditions, and operational contexts. The workshop will demonstrate how DTs enable real-time monitoring, predictive analysis, and what-if evaluation to support informed decision-making across physical and virtual domains. Through technical sessions and case studies, attendees will explore how DTs can be used for performance prediction, policy validation, fault diagnosis, resilience evaluation, and lifecycle management in complex IoT/IoV systems.

A core focus of the workshop is the **interplay between DTs and AI for network management**. Participants will learn how DT-generated data and models can be leveraged by AI and machine learning algorithms for training, inference, and control, and how AI can enhance DTs through model adaptation, behavior prediction, anomaly detection, and automated decision-making. This DT-AI synergy enables proactive and autonomous IoT/IoV operations, such as intelligent resource management, context-aware routing and scheduling, adaptive network slicing, cooperative perception and control, predictive maintenance, and resilient self-optimization. The workshop will further address key technical challenges in DT-AI integration, including scalable DT modeling and synchronization in highly mobile and partially observable environments, efficient data collection and fusion across heterogeneous IoT/IoV domains, latency- and

energy-constrained AI inference at the edge, and the trade-offs between communication, computation, and intelligence. Participants will be introduced to distributed and collaborative learning approaches, including privacy-preserving and resource-efficient AI techniques suitable for edge-cloud IoT/IoV deployments. In addition, the workshop will highlight practical considerations for building trustworthy and sustainable DT-AI-enabled systems, covering security, privacy, and reliability issues, as well as interoperability and standardization aspects relevant to IoT, IoV, and 6G ecosystems. Ethical and governance perspectives, such as bias mitigation and responsible AI deployment in shared DT environments, will also be discussed. By the end of the workshop, participants will be able to conceptualize and design **end-to-end DT-AI-enabled IoT/IoV solutions** that support predictive, self-optimizing, and resilient system operation. Researchers will gain insight into open research challenges and future directions, while practitioners will acquire practical knowledge for deploying intelligent IoT/IoV systems, positioning attendees at the forefront of next-generation smart mobility and intelligent IoT ecosystems.

V. Potential Audience.

This workshop will attract a multidisciplinary audience of researchers, engineers, and practitioners working at the intersection of digital twins, pervasive artificial intelligence, and next-generation IoT and Internet of Vehicles (IoV) systems. It is particularly relevant to professionals engaged in connected and autonomous vehicles, intelligent transportation systems, smart infrastructure, and large-scale IoT deployments. Automotive engineers, system architects, and original equipment manufacturers (OEMs) focused on connected, autonomous, and electric vehicles will benefit from insights into how digital twins and AI-enabled intelligence support reliable, low-latency, and adaptive IoV operations, including cooperative driving, traffic coordination, predictive maintenance, and vehicle-edge-cloud integration. These capabilities are essential for advancing safety-critical and data-intensive applications such as collision avoidance, cooperative perception, and intelligent fleet management.

Researchers and practitioners in wireless communications, networking, and 6G-enabled IoT/IoV architectures, including experts in mobility management, multi-connectivity, network slicing, edge computing, and ultra-reliable low-latency communications, will find value in discussions on DT-AI-driven system modeling, performance prediction, and autonomous resource optimization for highly dynamic and dense environments.

The workshop is also highly relevant to AI and machine learning researchers, data scientists, and system developers interested in pervasive intelligence for distributed cyber-physical systems. Topics such as DT-enhanced learning, edge-native and federated AI, cooperative decision-making, and anomaly detection will appeal to those developing intelligent, scalable, and trustworthy AI solutions across heterogeneous IoT/IoV platforms. Academics and graduate students in areas such as IoT, IoV, wireless communications, edge-cloud computing, cyber-physical systems, and network security will gain exposure to emerging research challenges, including AI-native architectures, DT-AI co-design, scalable system synchronization, and resilient intelligent operation.

Finally, industry innovators, startups, and technology providers working on smart mobility, intelligent infrastructure, and AI-driven IoT services will benefit from perspectives on translating DT-AI concepts into deployable systems, scalable platforms, and real-world testbeds. By fostering collaboration across these communities, the workshop aims to advance the development of predictive, autonomous, and resilient IoT/IoV ecosystems powered by digital twins and pervasive intelligence.

VI. Has the event previously been presented, if so, where? (100 word max)

Yes, we successfully organized two workshops in VTC-Spring 2025 and VTC-Fall 2025, respectively, with similar topics related to synergizing DTs and pervasive intelligence for next-generation IoV. This is the 3rd workshop we plan to organize in VTC-Spring 2026.

VII. Novelty.

This workshop introduces a **novel convergence of DTs and pervasive AI** to fundamentally rethink the design, operation, and optimization of next-generation IoT/IoV systems. It directly addresses key limitations of conventional IoT/IoV networking and control approaches, which often treat system modeling, protocol design, and AI-driven intelligence as loosely coupled or offline components. In contrast, the workshop advances a **holistic and tightly integrated DT-AI vision**, where DTs provide continuously updated virtual representations of physical devices, vehicles, networks, and environments, and AI enables autonomous, adaptive, and closed-loop system intelligence.

A core novelty lies in emphasizing **online and operational DTs** that interact bidirectionally with physical IoT/IoV systems. Rather than static models or simulation-only tools, DTs in this context evolve in real time by capturing system dynamics such as mobility patterns, network conditions, traffic load, service requirements, and operational contexts.

These DTs enable predictive insights, such as anticipating performance degradation, congestion buildup, or reliability risks, and support proactive decision-making when coupled with pervasive AI. This approach moves beyond reactive management toward **predictive and self-optimizing IoT/IoV operation**.

Another distinguishing feature of the workshop is its focus on **AI-native system intelligence across the full IoT/IoV stack**. Instead of using AI as an add-on analytics layer, the workshop highlights how DT-AI synergy can enable joint optimization of communication, computing, and control functions, including intelligent resource management, context-aware routing and scheduling, adaptive network slicing, cooperative perception, and edge computation offloading. Distributed, collaborative, and federated learning paradigms are emphasized to ensure scalability, low latency, and privacy preservation across heterogeneous and mobile IoT/IoV environments, overcoming the limitations of purely centralized AI solutions.

The workshop also foregrounds **trustworthy, resilient, and sustainable DT-AI-enabled systems** as a first-class research direction. DTs provide a structured substrate for incorporating security, reliability, and robustness into system intelligence, enabling anomaly detection, fault diagnosis, resilience assessment, and risk-aware decision-making in dynamic and open IoT/IoV ecosystems. Issues such as secure DT synchronization, robustness to data and model poisoning, and privacy-aware intelligence are treated as integral design concerns rather than afterthoughts.

By uniting these innovations, the workshop goes beyond incremental improvements and promotes a **paradigm shift toward autonomous, predictive, and resilient IoT/IoV systems**, where DTs and pervasive intelligence continuously co-design, validate, and operate complex cyber-physical infrastructures in real time. This integrated perspective positions the workshop as a distinctive forum for shaping the future of intelligent IoT, IoV, and smart mobility ecosystems in the 6G era.

VIII. List your Technical Program Committee.

Lian Zhao, Toronto Metropolitan University, Canada	Amr Salaheldin Hashem Matar, Huawei Canada, Canada
Khalid Aldubaikhy, Qassim University, Saudi Arabia	Weisen Shi, Huawei Canada, Canada
Sanaa Taha, Cairo University, Egypt	Peng Yang, Huazhong University of Science and Technology, China
Nizar H Alsharif, Al Baha University, Saudi Arabia	Tao Han, New Jersey Institute of Technology, USA
Wei Wang, Xian Jiaotong University, China	Takayuki Nishio, Tokyo Institute of Technology, Japan
Zubair Fadlullah, University of Western Ontario, Canada	Hai Jiang, Alberta University, Canada
Conghao Zhou, University of Waterloo, Canada	Ming Ding, Data61, Australia
Huaqing Wu, University of Calgary, Canada	Dinh Thai Hoang, University of Technology Sydney, Australia
Ruozhou Yu, North Carolina State University, USA	Yu Cheng, Illinois Institute of Technology, USA
Miao Wang, North Carolina State University, USA	Fang Fang, Western University Ontario, Canada
Zhi Liu, University of Electro-Communications, Japan	Mohamed M. E. A. Mahmoud, Tennessee Tech. University, USA
Nan Cheng, Xidian University, China	
Shaofeng Li, Peng Cheng Laboratory, China	
Shengbo Liu, Peng Chang Laboratory, China	

IX. A program outline.

Proposed Workshop Program Outline

The workshop is designed as a **half-day event** that combines keynote talks and peer-reviewed technical paper presentations, and interactive discussions to foster exchange between academia and industry on the synergy of **DTs and pervasive intelligence for next-generation IoT/IoV systems**.

1. Workshop Format and Paper Selection

The workshop will consist of **invited keynote sessions and technical paper presentations**. All submitted papers will undergo a rigorous peer-review process. We anticipate receiving approximately **20 paper submissions**, from which **8 high-quality papers** will be selected for oral presentation, corresponding to an expected acceptance rate of **40%**. Accepted papers will reflect cutting-edge research aligned with the workshop’s themes, including DT-AI integration, IoT/IoV architectures, edge-cloud intelligence, and 6G-enabled systems.

2. Participants and Discussion Scope

The workshop will bring together **leading researchers and engineers from academia and industry**, as well as authors of accepted papers. Interactive discussions will be an integral part of the program and will focus on a set of **key open research questions and emerging challenges**, such as:

- Digital twins for large-scale IoT and IoV systems
- AI-native architectures for intelligent IoT/IoV networking
- DT-AI synergy for real-time optimization and autonomous control
- Edge-cloud intelligence and distributed learning for IoT/IoV

- Trustworthy, secure, and privacy-preserving DT-AI systems
- Applications of foundation models and large language models (LLMs) in DT-enabled IoT/IoV
- Standardization, interoperability, and evaluation frameworks for DT-AI-enabled systems
- Practical deployment experiences, testbeds, and field trials

3. Review Process

Each paper submission will be managed by a **TPC Co-Chair without conflict of interest** and assigned to **at least three independent reviewers**. Based on reviewer feedback and scores, the TPC will rank submissions and select accepted papers through a transparent and fair evaluation process.

4. Expected Participation and Outreach

The Call for Papers (CFP) will be widely disseminated through relevant research and professional communities, including IoT, IoV, wireless communications, networking, AI, and intelligent transportation systems. The CFP will be actively promoted by our Publicity Co-chairs and our TPC team across different regions worldwide. Given the timeliness and interdisciplinary nature of the topic, along with the participation of distinguished speakers, we anticipate **20-30 attendees**.

5. Tentative Workshop Schedule (Half-Day)

- **08:30 - 08:35** Welcome and Opening Remarks
- **08:35 - 09:10** Keynote Talk I
- **09:10 - 10:10** Technical Session I (4 technical presentations)
- **10:10 - 10:20** Coffee Break
- **10:20 - 10:55** Keynote Talk II (4 Oral Presentations)
- **10:55 - 11:55** Technical Session II (4 technical presentations)
- **11:55 - 12:00** Closing Remarks

**IEEE VTC-Spring 2026: The 3rd Workshop on
Synergizing Digital Twins and Pervasive Intelligence for Next-Generation IoT/IoV
Call for Papers**

Next-generation Internet of Things (IoT) and Internet of Vehicles (IoV) systems are evolving toward highly autonomous, hyper-connected, and intelligence-driven ecosystems. Future networking, computing, and control infrastructures must support massive-scale connectivity, high mobility, heterogeneous services, and ultra-reliable low-latency communications across diverse IoT/IoV scenarios, including connected and autonomous vehicles, intelligent transportation systems, roadside and aerial sensing, and edge-cloud collaborative environments. These systems face fundamental challenges arising from dynamic and large-scale network topologies, rapidly time-varying wireless conditions, intermittent connectivity, spectrum scarcity, ultra-dense deployments, and stringent safety, reliability, and latency requirements for mission-critical applications. Digital twins (DTs) have emerged as a powerful paradigm for managing the complexity of IoT/IoV systems by providing virtual representations of physical entities, networks, and environments that continuously reflect system states and dynamics. When combined with pervasive artificial intelligence (AI), embedded across devices, vehicles, edge nodes, and cloud platforms, DTs enable predictive analysis, autonomous decision-making, and closed-loop optimization across physical and virtual domains. This DT-AI synergy supports proactive and self-optimizing IoT/IoV operation, including intelligent resource management, context-aware routing and scheduling, adaptive network slicing, cooperative perception and control, predictive maintenance, and resilient system adaptation envisioned for 6G-enabled IoT/IoV ecosystems. Despite their promise, integrating DTs and AI in large-scale and highly dynamic IoT/IoV systems raises significant challenges, such as scalable DT modeling and synchronization under mobility and partial observability, efficient data collection and fusion across heterogeneous domains, latency- and energy-constrained AI inference at the edge, and ensuring trustworthy, secure, and privacy-preserving DT-AI operations. Addressing these challenges requires new architectures, protocols, and cross-layer optimization frameworks that jointly consider sensing, communication, computing, and intelligence.

This workshop invites high-quality original research contributions that explore theories, architectures, algorithms, and experimental systems for synergizing digital twins and pervasive intelligence in next-generation IoT/IoV. Both theoretical advances and practical insights from testbeds, prototypes, and field trials are encouraged.

- DT architectures and modeling techniques for IoT and IoV systems
- AI-enhanced digital twins for prediction, optimization, and autonomous control
- Joint DT-AI frameworks for real-time IoT/IoV network and system optimization
- Pervasive intelligence in self-organizing, adaptive, and autonomous IoT/IoV systems
- Edge-cloud collaboration for DT-enabled intelligence in IoT/IoV
- Distributed, federated, and privacy-preserving learning for DT-AI integration
- Resource management, routing, scheduling, and network slicing enabled by DT-AI
- Security, privacy, and trust in DT-AI-enabled IoT/IoV ecosystems
- Interoperability and standardization of DT and AI technologies for IoT/IoV and 6G
- Big data management and real-time analytics for large-scale digital twins
- DT-AI-enabled applications in intelligent transportation and smart infrastructure
- UAV/drone-assisted IoT/IoV sensing and monitoring using DT and AI
- Experimental platforms, testbeds, and field trials for DT-AI-enabled IoT/IoV
- Cross-domain innovations enabled by DT-AI synergy in smart mobility and beyond

Paper submission: Papers must be formatted in the standard IEEE two-column format that is used by the VTC-Spring 2026 main conference, and must not exceed six pages in length (including references).

Workshop Organization Committee

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Mingzhe Chen, University of Miami, USA

Technical Program Co-chairs

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Qihao Li, Jilin University, China
Yuchen Liu, North Carolina State University, USA
Katsuya Suto, University of Electro-Comm, Japan
Mushu Li, Lehigh University, USA
Hong Chen, University of New Brunswick, Canada
Haixia Peng, Xi'an Jiaotong University, China

Publicity Co-chairs

Jie Gao, Carleton University, Canada
Nan Chen, University of Ottawa, Canada
Omar Alhussein, Khalifa University, UAE

Important Dates:

Paper Submission Deadline: 25 January 2026
Paper Acceptance Notification: 18 March 2026
Final Paper Submission: 1 April 2026

